Chapter 1

Matrices

1.1 Matrices and Linear Maps

Matrices are the "coordinated" way to describe linear transformations. We will formalise this in Theorem 2.3 below. But first, we define a linear transformation.

Definition 1.1 (Linear transformation)**.** Fix a field F and two vector spaces V, W over F. A *linear transformation* (or linear map) is a function $T : V \to W$ such that:

- **Additivity:** $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
- **Homogeneity:** $T(\lambda v) = \lambda T(v)$ for all $\lambda \in F$ and $v \in V$.

Lemma 1.2. *If $T : V \to W$ is a linear transformation, then $T(0) = 0$.*

Proof. By linearity, $T(0) = T(0 + 0) = T(0) + T(0)$. Adding $-T(0)$ (the additive inverse of $T(0)$ in W) to both sides yields $T(0) = 0$. □

Recall that any finite-dimensional vector space V, with dimension n, is isomorphic to F^n. For now, when we say we have written V and W in coordinates, we mean we replace V and W with F^n and F^m. This motivates the concept of matrices.

Theorem 1.3. *For all linear maps $T : F^n \to F^m$ defined by $(x_1, \ldots, x_n) \mapsto (y_1, \ldots, y_m)$, there exists an array of numbers (a_{ij}) such that:*

$$y_i = \sum_{j=1}^{n} a_{ij} x_j$$

Proof. We define the standard elementary vectors $e_j \in F^n$ to have value 1 in the j-th entry, and 0 everywhere else. Since $T(e_j)$ is a vector in F^m, we can write it as a list of m coordinates. For all j, define $a_{1j}, \ldots, a_{mj} \in F$ by:

$$T(e_j) = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

(Idea: think of $T(e_j)$ as the j^{th} column of the matrix (a_{ij})).

Any input vector can be written as $(x_1, \ldots, x_n) = x_1 e_1 + \cdots + x_n e_n$ by our definition of addition in F^n. By the linearity of T, we have:

$$\begin{aligned} T(x_1, \ldots, x_n) &= T(x_1 e_1 + \cdots + x_n e_n) \\ &= x_1 T(e_1) + \cdots + x_n T(e_n) \\ &= \sum_{j=1}^{n} x_j \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} \end{aligned}$$

Taking the i^{th} row (coordinate) of the above sum, we get:

$$y_i = \sum_{j=1}^{n} a_{ij}\, x_j$$

as required. □

Remark. The converse is also true: any function $T : F^n \to F^m$ given by such an array (a_{ij}) is a linear map. This is fundamental: *linear maps are exactly the functions that can be represented by matrices.*

Observe that all the information needed to compute this linear transformation is encoded in the array $A \coloneqq (a_{ij})$.

Definition 1.4 (Matrix)**.** The set of $m \times n$ matrices over a field F is defined as:

$$M_{m\times n}(F) \coloneqq \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} : a_{ij} \in F \right\}$$

How do we conveniently encode the action of T using this? We define matrix-vector multiplication:

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

The upshot is that, given a linear map $T : F^n \to F^m$, we *define* matrix multiplication with a column vector precisely to execute this map.

1.2 Matrix Multiplication

1.2.1 Motivation

First, we motivate the definition of general matrix multiplication. Consider two linear maps:

- $T : F^k \to F^n$ represented by the matrix $B \in M_{n\times k}(F)$
- $S : F^n \to F^m$ represented by the matrix $A \in M_{m\times n}(F)$

Define the composite linear map $R : F^k \to F^m$ by $R = S \circ T$. By Claim ($\star$), this new linear map must also have a matrix representation, which we will call $C \in M_{m\times k}(F)$.

We *want* to calculate $c_{i\ell}$, the $(i,\ell)^{\text{th}}$ element of the matrix C. Let's calculate the ℓ^{th} column of C, call it C_ℓ. We know that each column of C represents the effect of the linear map R on the standard basis vector e_ℓ:

$$C_\ell = R(e_\ell) = S(T(e_\ell))$$

We know $T(e_\ell)$ is exactly the ℓ^{th} column of B, call it B_ℓ. So we must calculate $S(B_\ell)$. Because B_ℓ is a column vector in F^n, evaluating $S(B_\ell)$ is equivalent to taking a linear combination of the columns of A (which we will denote $A_1, \dots, A_n$):

$$S(B_\ell) = b_{1\ell}A_1 + \cdots + b_{n\ell}A_n$$

To find $c_{i\ell}$, we take the i^{th} row of this result:

$$\begin{aligned} c_{i\ell} = (C_\ell)_i &= b_{1\ell}(A_1)_i + \cdots + b_{n\ell}(A_n)_i \\ &= b_{1\ell}a_{i1} + \cdots + b_{n\ell}a_{in} \\ &= \sum_{j=1}^{n} a_{ij}b_{j\ell} \end{aligned}$$

This is exactly the formula for $c_{i\ell}$ that we use to define matrix multiplication. Therefore, the fact that composition of linear maps gives the formula above for the numbers $c_{i\ell}$ is why we define matrix multiplicaation in the following way:

Definition 1.5 (Matrix operations)**.**

- Matrix multiplication is the function $\cdot : M_{m\times n}(F) \times M_{n\times p}(F) \to M_{m\times p}(F)$ defined by:

$$(a_{ij})_{m\times n}\,(b_{k\ell})_{n\times p} := \left(c_{i\ell} = \sum_{j=1}^{n} a_{ij}b_{j\ell}\right)_{m\times p}$$

- Matrix addition is the function $+ : M_{m\times n}(F) \times M_{m\times n}(F) \to M_{m\times n}(F)$ defined by:

$$(a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij}).$$

- Scalar multiplication for matrices is the function $\cdot : F \times M_{m\times n}(F) \to M_{m\times n}(F)$ defined by:

$$\lambda(a_{ij}) = (\lambda a_{ij}).$$

Remark. Just like $F^n \cong M_{n\times 1}(F)$, the set $M_{m\times n}(F)$ forms a vector space under matrix addition and scalar multiplication. However, matrix multiplication gives $M_{n\times n}(F)$ *additional* structure beyond just a vector space (it forms an algebra).

Definition 1.6 (Square Matrix)**.** A matrix is square if $A \in M_{n\times n}(F)$ (the number of rows equals the number of columns). Square matrices of a fixed size can be multiplied by each other.

Proposition 1.7. *Matrix multiplication is associative:* $(AB)C = A(BC)$.

Proof.

$$((AB)C)_{i\ell} = \sum_j (AB)_{ij} c_{j\ell} = \sum_j \left(\sum_k a_{ik} b_{kj} \right) c_{j\ell} = \sum_{j,k} a_{ik} b_{kj} c_{j\ell}$$

$$(A(BC))_{i\ell} = \sum_k a_{ik} (BC)_{k\ell} = \sum_k a_{ik} \left(\sum_j b_{kj} c_{j\ell} \right) = \sum_{j,k} a_{ik} b_{kj} c_{j\ell}$$

Therefore, $((AB)C)_{i\ell} = (A(BC))_{i\ell}$. □

Corollary 1.8. *Given linear maps* $S : F^m \to F^n$ *and* $T : F^n \to F^p$ *with matrix representations* A *and* B *(so* $S(v) = Av$ *and* $T(v) = Bv$*), the matrix representation of the composite map* $T \circ S : F^m \to F^p$ *is given by the matrix product* BA*.*

Proof. $T \circ S(v) = T(S(v)) = B(Av) = (BA)v$ by associativity. □

1.2.2 Some more formal properties of matrices

Definition 1.9 (Bilinear map)**.** Let V, W, U be vector spaces over F. A map $f : V \times W \to U$ mapping $(v, w) \mapsto f(v, w)$ is *bilinear* if it is linear in each argument when the other is held fixed:

- $(\lambda_1 v_1 + \lambda_2 v_2, w) \mapsto \lambda_1 f(v_1, w) + \lambda_2 f(v_2, w)$
- $(v, \lambda_1 w_1 + \lambda_2 w_2) \mapsto \lambda_1 f(v, w_1) + \lambda_2 f(v, w_2)$

Remark. Note that this implies the standard scalar extraction property $f(\lambda v, w) = f(v, \lambda w) = \lambda f(v, w)$ (simply set $\lambda_2 = 0$).

Example 1.10. The dot product $F^n \times F^n \to F$, defined by $((a_1, \ldots, a_n), (b_1, \ldots, b_n)) \mapsto a_1 b_1 + \cdots + a_n b_n$ is bilinear. This follows easily from the distributive properties of the field F.

We want to prove that matrix multiplication is bilinear. To do this, we first establish how matrix multiplication interacts with columns and rows.

Proposition 1.11. *Let* $A \in M_{m\times n}(F)$ *and* $B = (B_1, \ldots, B_p) \in M_{n\times p}(F)$*, where* B_j *are the columns of* B*. Then:*

$$AB = (AB_1, \ldots, AB_p)$$

Proof. By the formula, $(AB)_{i\ell} = \sum_k a_{ik} b_{k\ell}$. Notice that $\sum_k a_{ik} b_{k\ell}$ is exactly the i^{th} element of the matrix-vector product AB_ℓ. Thus, the ℓ^{th} column of AB is exactly the vector AB_ℓ. □

Remark. This provides a *column-wise* way to calculate matrix products. To find the entire j^{th} column of AB, just calculate the matrix-vector product AB_j. The j^{th} column of AB **only** depends on the j^{th} column of B.

Proposition 1.12. *We can separate rows under right multiplication. Let B_i be the rows of B. Then:*

$$\begin{pmatrix} B_1 \\ \vdots \\ B_p \end{pmatrix} A = \begin{pmatrix} B_1 A \\ \vdots \\ B_p A \end{pmatrix}$$

Proof. Same logic as above, switching indices appropriately. □

Corollary 1.13. *Matrix multiplication is bilinear.*

Proof. We prove linearity in the second argument. Let A be an $m \times n$ matrix, B and C be $n \times p$ matrices, and $\lambda \in F$.

$$\begin{aligned} A(\lambda B) &= A(\lambda B_1 \dots \lambda B_n) \\ &= (A(\lambda B_1) \dots A(\lambda B_n)) && \text{(Column separation)} \\ &= (\lambda(AB_1) \dots \lambda(AB_n)) && \text{(Matrix-vector linearity)} \\ &= \lambda(AB_1 \dots AB_n) \\ &= \lambda(AB) \end{aligned}$$

And for addition:

$$\begin{aligned} A(B + C) &= A(B_1 + C_1 \dots B_n + C_n) \\ &= (A(B_1 + C_1) \dots A(B_n + C_n)) \\ &= (A(B_1) + A(C_1) \dots A(B_n) + A(C_n)) \\ &= (A(B_1) \dots A(B_n)) + (A(C_1) \dots A(C_n)) \\ &= AB + AC \end{aligned}$$

Linearity in the first argument follows similarly. □

Definition 1.14 (Transpose)**.** The transpose of a matrix $A = (a_{ij}) \in M_{m\times n}(F)$ is defined as $A^\top := (a_{ji}) \in M_{n\times m}(F)$.

Proposition 1.15. *The transpose operation satisfies the following properties:*

- *Transpose is linear: $(A + B)^\top = A^\top + B^\top$ and $(\lambda A)^\top = \lambda A^\top$.*
- $(A^\top)^\top = A$.
- $(AB)^\top = B^\top A^\top$.

Proof. We prove the product rule. Let $C = AB$. Then $c_{ij} = \sum_k a_{ik}b_{kj}$. The (i, j) entry of $(AB)^\top$ is the (j, i) entry of AB, which is $\sum_k a_{jk}b_{ki}$. The (i, j) entry of $B^\top A^\top$ is the dot product of the i-th row of $B^\top$ and the j-th column of $A^\top$. This is exactly $\sum_k (B^\top)_{ik}(A^\top)_{kj} = \sum_k b_{ki}a_{jk} = \sum_k a_{jk}b_{ki}$. □

Remark. Notice that the order reverses in $(AB)^\top = B^\top A^\top$. This is necessary for the dimensions to align. For example, if A is 1×5 and B is 5×1, AB is 1×1. $A^\top$ is 5×1 and $B^\top$ is 1×5. The product $A^\top B^\top$ is impossible, but $B^\top A^\top$ results in a 1×1 matrix. There is a deeper reason for this, which we will see when we look at dual vector spaces and dual linear transformations.

1.2.3 Special Matrices

Definition 1.16 (Symmetric Matrix)**.** A square matrix A is *symmetric* if $A = A^\top$.

Definition 1.17 (Orthogonal Matrix)**.** A square matrix A is *orthogonal* if $AA^\top = I$.

Remark. For a square matrix, $AA^\top = I \iff A^\top A = I$. What does it mean for $AA^\top = I$? We will discuss this in more detail in a later chapter, but for now it is enough to note that if A_i are the rows of A, then $\left(AA^\top\right)_{ij} = A_i \cdot A_j$. Therefore:

$$A_i \cdot A_j = \delta_{ij} \coloneqq \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

This means the *rows* of A are orthonormal. Similarly, $A^\top A = I$ implies the *columns* of A are orthonormal. Geometrically in $\mathbb{R}^n$, multiplying by an orthogonal matrix represents a rigid transformation (a rotation or reflection) that preserves lengths and angles (again, this will make more sense in a later chapter).

Definition 1.18 (Triangular and Diagonal Matrices)**.** Let $A = (a_{ij})$ be a matrix.

- A is *upper-triangular* if $a_{ij} = 0$ for all $i > j$.
- A is *lower-triangular* if $a_{ij} = 0$ for all $i < j$.
- A is *diagonal* if it is both upper and lower triangular (i.e., $a_{ij} = 0$ for all $i \neq j$).